Class 10 Maths — Complete Cheat Sheet

CBSE · Every formula, theorem and method from the basics up to the board exam · with worked examples and full proofs · **Part A** foundations (Classes 6–9) · **Part B** all 14 Class 10 chapters · **Part C** exam plan

1 Real Numbers | 2 Polynomials | 3 Linear Equations | 4 Quadratic Equations | 5 Arithmetic Progressions | 6 Triangles | 7 Coordinate Geometry | 8 Trigonometry | 9 Heights & Distances | 10 Circles | 11 Areas Related to Circles | 12 Surface Areas & Volumes | 13 Statistics | 14 Probability

Items marked **EXTRA** are outside the current rationalised Class 10 syllabus, but are short, useful and often needed in later classes or competitive exams.

Part A — Foundations you must already own

Classes 6–9

Types of numbers

- **Natural (N):** 1, 2, 3 … **Whole (W):** 0, 1, 2 …
- **Integers (Z):** … –2, –1, 0, 1, 2 …
- **Rational (Q):** can be written $\frac{p}{q}$, $q \neq 0$. Decimal form *terminates* or *repeats* (0.75, 0.333…).
- **Irrational:** decimal never ends, never repeats (√2, √3, π).
- **Real (R):** rational + irrational — every point on the number line.
- **Prime:** exactly 2 factors (2, 3, 5, 7, 11 …). 2 is the only even prime. **1 is neither prime nor composite.**
- **Co-prime:** HCF = 1 (e.g. 8 and 15).

Order of operations and signs

BODMAS: Brackets → Orders (powers, roots) → Division → Multiplication → Addition → Subtraction.

(+)(+) = +, (–)(–) = +, (+)(–) = –. –(a – b) = –a + b.

Watch out: $(-3)^2 = 9$ but $-3^2 = -9$. Division by 0 is not defined.

Divisibility rules

By	Rule
2	last digit even
3	sum of digits divisible by 3
4	last two digits divisible by 4
5	last digit 0 or 5
6	divisible by both 2 and 3
8	last three digits divisible by 8
9	sum of digits divisible by 9
10	last digit 0
11	(sum of odd-place digits) – (sum of even-place digits) is 0 or a multiple of 11

Fractions, percentages, ratio

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd} \qquad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \qquad \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

$$x\% \text{ of } y = \frac{x}{100} \times y \qquad \%\ \text{change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

½ = 50% · ⅓ = 33.33% · ¼ = 25% · ⅕ = 20% · ⅛ = 12.5% · ¾ = 75%

Proportion: a : b = c : d ⇒ ad = bc. **Direct:** $\frac{x}{y}$ = constant. **Inverse:** xy = constant.

Laws of exponents and surds

$$a^m \cdot a^n = a^{m+n} \qquad a^m \div a^n = a^{m-n} \qquad (a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n \qquad a^0 = 1 \qquad a^{-n} = \frac{1}{a^n} \qquad a^{1/n} = \sqrt[n]{a}$$

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{(ab)} \qquad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{(a/b)} \qquad (\sqrt{a})^2 = a$$

Rationalising: $\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$ (multiply top and bottom by the conjugate).

Watch out: $\sqrt{(a+b)} \neq \sqrt{a} + \sqrt{b}$. $(a+b)^2 \neq a^2 + b^2$.

Algebraic identities (learn by heart)

$$(a+b)^2 = a^2 + 2ab + b^2$$
$$(a-b)^2 = a^2 - 2ab + b^2$$
$$a^2 - b^2 = (a+b)(a-b)$$
$$(x+a)(x+b) = x^2 + (a+b)x + ab$$
$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$
$$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$$
$$(a-b)^3 = a^3 - b^3 - 3ab(a-b)$$
$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$
$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$
$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

If a + b + c = 0, then $a^3 + b^3 + c^3 = 3abc$.

Remainder theorem: p(x) ÷ (x – a) leaves remainder p(a). **Factor theorem:** (x – a) is a factor ⇔ p(a) = 0.

Everyday arithmetic formulas

$$\text{Profit \%} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100 \qquad \text{Discount} = \text{MP} - \text{SP}$$

$$\text{Simple interest} = \frac{\text{P} \times \text{R} \times \text{T}}{100}$$

$$\text{Compound interest: } \text{A} = \text{P}\left(1 + \frac{\text{R}}{100}\right)^{\text{n}}$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$1\ \text{km/h} = \frac{5}{18}\ \text{m/s} \qquad 1\ \text{m/s} = \frac{18}{5}\ \text{km/h}$$

Work: if A finishes in *a* days, A does $\frac{1}{a}$ of the work per day.

Together: $\frac{1}{a} + \frac{1}{b}$ per day.

Values worth memorising

$\sqrt{2} = 1.414 \cdot \sqrt{3} = 1.732 \cdot \sqrt{5} = 2.236 \cdot \pi = \frac{22}{7} \approx 3.14$

Squares: 11^2 121 · 12^2 144 · 13^2 169 · 14^2 196 · 15^2 225 · 16^2 256 · 17^2 289 · 18^2 324 · 19^2 361 · 20^2 400 · 21^2 441 · 22^2 484 · 23^2 529 · 24^2 576 · 25^2 625

Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000 (1^3 to 10^3)

Pythagorean triples: 3-4-5 · 5-12-13 · 8-15-17 · 7-24-25 · 9-40-41 · 20-21-29 (and their multiples, e.g. 6-8-10)

Lines and angles

- Complementary angles add to 90°; supplementary add to 180°.
- Linear pair = 180°. Vertically opposite angles are equal.
- **Parallel lines cut by a transversal:** corresponding angles equal · alternate interior angles equal · co-interior angles add to 180°.
- Polygon with n sides: sum of interior angles = (n − 2) × 180°. Regular polygon: each exterior angle = $\frac{360°}{n}$.

Triangles — basic facts

- Angle sum = 180°. Exterior angle = sum of the two interior opposite angles.
- Sum of any two sides > third side. Bigger angle faces the longer side.
- Isosceles: angles opposite equal sides are equal. Equilateral: each angle 60°.
- **Congruence rules:** SSS · SAS · ASA · AAS · RHS. (No "SSA" or "AAA" for congruence.)
- **Pythagoras:** in a right triangle, Hypotenuse² = Base² + Perpendicular². Converse: if $c^2 = a^2 + b^2$, the angle opposite c is 90°.
- **Mid-point theorem:** the line joining mid-points of two sides is parallel to the third side and half of it.

Quadrilaterals (angle sum 360°)

Shape	Key properties
Parallelogram	opposite sides equal and parallel; opposite angles equal; diagonals bisect each other
Rectangle	parallelogram + all angles 90° + diagonals equal
Rhombus	parallelogram + all sides equal + diagonals bisect at 90°
Square	rectangle + rhombus (everything above)
Trapezium	one pair of opposite sides parallel

Circle theorems from Class 9

- Perpendicular from the centre to a chord bisects the chord.
- Equal chords are equidistant from the centre (and vice versa).
- Angle at the centre = 2 × angle at the circumference on the same arc.
- Angles in the same segment are equal. Angle in a semicircle = 90°.
- Cyclic quadrilateral: opposite angles add to 180°.

Mensuration — plane figures

Figure	Area	Perimeter
Rectangle	$l \times b$	$2(l + b)$
Square	a^2 (diagonal $a\sqrt{2}$)	$4a$
Triangle	½ × base × height	$a + b + c$
Triangle (Heron)	$\sqrt{s(s-a)(s-b)(s-c)}$	$s = \frac{a+b+c}{2}$
Equilateral	$\frac{\sqrt{3}}{4} a^2$ (height $\frac{\sqrt{3}}{2} a$)	$3a$
Parallelogram	base × height	$2(a + b)$
Rhombus	½ × $d_1 \times d_2$	$4a$
Trapezium	½ $(a + b) \times h$	sum of sides
Circle	πr^2	$2\pi r$

Mensuration — solids

Solid	Curved / lateral	Total surface	Volume
Cuboid	$2h(l + b)$	$2(lb + bh + hl)$	lbh
Cube	$4a^2$	$6a^2$	a^3
Cylinder	$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Cone	πrl	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

Cone slant height $l = \sqrt{(r^2 + h^2)}$. Cuboid diagonal = $\sqrt{(l^2 + b^2 + h^2)}$. Cube diagonal = $a\sqrt{3}$.

Units: 1 m^3 = 1000 litres · 1 litre = 1000 cm^3 · 1 m^2 = 10 000 cm^2 · 1 hectare = 10 000 m^2.

Solving a linear equation in one variable

- Whatever you do to one side, do to the other. Moving a term across "=" changes its sign; a multiplier becomes a divisor.
- Clear fractions first by multiplying every term by the LCM of the denominators.
- **Cross-multiplication:** $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$.

$$3x - 7 = 2x + 5 \Rightarrow 3x - 2x = 5 + 7 \Rightarrow x = 12$$

$$\frac{2x+1}{3} = \frac{x-1}{2} \Rightarrow 4x + 2 = 3x - 3 \Rightarrow x = -5$$

Tip: Put your answer back in the original equation. LHS must equal RHS.

Factorising — four methods

1. **Common factor:** $6x^2 + 9x = 3x(2x + 3)$.
2. **Grouping:** $ax + ay + bx + by = (a + b)(x + y)$.
3. **Identities:** $4x^2 - 25 = (2x + 5)(2x - 5)$; $x^2 + 6x + 9 = (x + 3)^2$.
4. **Splitting the middle term** of $ax^2 + bx + c$: find two numbers with product ac and sum b.

$$x^2 - 5x + 6 \to \text{product } 6, \text{sum } -5 \to -2, -3 \to (x - 2)(x - 3)$$

$$6x^2 + 7x - 3 \to \text{product } -18, \text{sum } 7 \to 9, -2$$
$$= 6x^2 + 9x - 2x - 3 = 3x(2x + 3) - 1(2x + 3)$$
$$= (3x - 1)(2x + 3)$$

HCF, LCM and square roots by prime factors

Break each number into primes using a factor tree or repeated division by 2, 3, 5, 7 …

$$72 = 2^3 \times 3^2 \quad 120 = 2^3 \times 3 \times 5$$
$$\text{HCF} = 2^3 \times 3 = 24 \quad \text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

Square root: pair up equal primes and take one from each pair. $\sqrt{7056} = \sqrt{(2^4 \times 3^2 \times 7^2)} = 2^2 \times 3 \times 7 = 84$.

Fractions: LCM of fractions = $\frac{\text{LCM of numerators}}{\text{HCF of denominators}}$; HCF of fractions = $\frac{\text{HCF of numerators}}{\text{LCM of denominators}}$.

Linear equations in two variables and graphs (Class 9)

- $ax + by + c = 0$ has infinitely many solutions; its graph is a straight line. Every point on the line is a solution.
- To plot: choose 3 values of x, find y, plot and join. Two points fix the line; the third is a check.
- x-intercept: put y = 0. y-intercept: put x = 0.
- $x = a$ is a vertical line (parallel to the y-axis). $y = b$ is a horizontal line (parallel to the x-axis).
- x-axis is the line $y = 0$; y-axis is the line $x = 0$. The origin is (0, 0).
- In (x, y): x is the abscissa (distance from the y-axis), y is the ordinate (distance from the x-axis).

Ungrouped data: mean, median, mode

$$\text{Mean} = \frac{\text{sum of observations}}{\text{number of observations}}$$

Median: arrange in order. If n is odd, it is the $(\frac{n+1}{2})$th value. If n is even, it is the average of the $(\frac{n}{2})$th and $(\frac{n}{2} + 1)$th values.

Mode: the value that occurs most often. **Range:** highest – lowest.

Data 3, 7, 7, 9, 14: mean = 40 ÷ 5 = 8, median = 7, mode = 7, range = 11.

- If every value increases by k, the mean increases by k. If every value is multiplied by k, the mean is multiplied by k.
- The sum of deviations from the mean is always 0.

Unit conversions

Length	Area	Volume / capacity
1 km = 1000 m	1 m² = 10 000 cm²	1 m³ = 1 000 000 cm³
1 m = 100 cm	1 cm² = 100 mm²	1 m³ = 1000 litres = 1 kilolitre
1 cm = 10 mm	1 hectare = 10 000 m²	1 litre = 1000 cm³ = 1000 ml
1 inch = 2.54 cm	1 km² = 100 hectares	1 cm³ = 1 ml

Watch out: When length units change by a factor k, area changes by k^2 and volume by k^3.

The language of proofs

- **Axiom / postulate:** accepted without proof. **Theorem:** a statement that has been proved.
- **Converse:** swap "if" and "then". The converse of a true statement need not be true.
- **Proof by contradiction:** assume the opposite, reach something impossible, conclude the original is true.
- **Counter-example:** one example is enough to show a statement is false, but examples never prove it true.
- Geometry proofs are written as: Given → To prove → Construction → Proof, with a reason for each step (CPCT, AA, alternate angles …).

Example · **Heron's formula**

Triangle with sides 13 cm, 14 cm, 15 cm.

$$s = \frac{13 + 14 + 15}{2} = 21$$
$$\text{Area} = \sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7056} = 84 \text{ cm}^2$$

Altitude on the 14 cm side = $\frac{2 \times 84}{14} = 12$ cm.

Mental-maths shortcuts

- **Square of a number ending in 5:** $35^2 \to 3 \times 4 = 12$, attach 25 → 1225.
- **Multiply by 11:** 43 × 11 → 4 (4 + 3) 3 → 473.
- **Multiply by 5:** halve, then × 10. **By 25:** divide by 4, then × 100.
- **Difference of squares:** $47 \times 53 = 50^2 - 3^2 = 2491$.
- **Near-100 squares:** $98^2 = (100 - 2)^2 = 10000 - 400 + 4 = 9604$.
- $1/\sqrt{2} \approx 0.707 \cdot 1/\sqrt{3} \approx 0.577 \cdot \sqrt{3}/2 \approx 0.866$.

Part B — Class 10, chapter by chapter

1 · Real Numbers

Unit: Number Systems · 6 marks

Fundamental Theorem of Arithmetic

Every composite number can be written as a product of primes, and this factorisation is **unique** apart from the order of the factors. Example: $360 = 2^3 \times 3^2 \times 5$.

HCF = product of the *smallest* power of each *common* prime.
LCM = product of the *greatest* power of *every* prime involved.

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

Watch out: This product rule works for two numbers only, not three.

Tip: "Largest number that divides…" → HCF. "Smallest number divisible by…" or "when will they meet / ring together again" → LCM.

Proving irrationality (√2, √3, √5)

Key theorem: if a prime p divides a^2, then p divides a.

Proof pattern (contradiction):

- Assume $\sqrt{2} = \frac{a}{b}$, with a, b co-prime, $b \neq 0$.
- Square: $a^2 = 2b^2 \Rightarrow 2$ divides $a^2 \Rightarrow 2$ divides a. Let $a = 2c$.
- Then $4c^2 = 2b^2 \Rightarrow b^2 = 2c^2 \Rightarrow 2$ divides b.
- So a and b share the factor 2 — contradicts co-prime. Hence √2 is irrational.

For 3 + 2√5 type: assume it is rational, isolate √5 = rational → contradiction.

Rational ± irrational = irrational. Non-zero rational × irrational = irrational.

Standard question types

- **Can 6^n end in 0?** A number ending in 0 needs prime factors 2 *and* 5. $6^n = 2^n \times 3^n$ has no 5 → never.
- **Show 7 × 11 × 13 + 13 is composite:** take 13 common → 13 × (78), more than two factors.

EXTRA **Euclid's lemma:** $a = bq + r$, $0 \leq r < b$. **Decimals:** $\frac{p}{q}$ (lowest terms) terminates only if $q = 2^m \times 5^n$.

Example · HCF and LCM by prime factorisation

Find the HCF and LCM of 96 and 404 and verify the product rule.

$$96 = 2^5 \times 3 \quad 404 = 2^2 \times 101$$
$$\text{HCF} = 2^2 = 4 \quad \text{LCM} = 2^5 \times 3 \times 101 = 9696$$
$$4 \times 9696 = 38\,784 = 96 \times 404 \checkmark$$

Given the HCF, find the LCM: HCF(306, 657) = 9 ⇒ LCM = $\frac{306 \times 657}{9}$ = 22 338.

Example · Three numbers

HCF and LCM of 6, 72, 120.

$$6 = 2 \times 3 \quad 72 = 2^3 \times 3^2 \quad 120 = 2^3 \times 3 \times 5$$
$$\text{HCF} = 2 \times 3 = 6 \quad \text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

Here HCF × LCM = 2160, but 6 × 72 × 120 = 51 840. The product rule fails for three numbers.

Example · Word problems

- **Meeting again:** Sonia takes 18 min and Ravi 12 min for one round of a field, starting together. They meet at the start after LCM(18, 12) = 36 minutes.
- **Bells:** bells ring every 9, 12 and 15 minutes. LCM = 180 minutes, so they ring together every 3 hours.
- **Remainder type:** largest number that divides 245 and 1029 leaving remainder 5 in each case = HCF(245 − 5, 1029 − 5) = HCF(240, 1024) = 16.
- **Largest equal groups / longest tape / biggest tile:** HCF.

Proof · **5 – √3 is irrational**

1. Assume $5 - \sqrt{3}$ is rational. Then $5 - \sqrt{3} = \frac{a}{b}$, where a, b are co-prime integers and $b \neq 0$.
2. Rearranging: $\sqrt{3} = 5 - \frac{a}{b} = \frac{5b - a}{b}$.
3. Since a and b are integers, $\frac{5b - a}{b}$ is rational, so $\sqrt{3}$ is rational.
4. But $\sqrt{3}$ is irrational. This is a contradiction.
5. So the assumption is wrong, and $5 - \sqrt{3}$ is irrational.

The same five lines work for $3 + 2\sqrt{5}$, $7\sqrt{5}$, $\frac{1}{\sqrt{2}}$ and similar numbers. If asked, first prove $\sqrt{3}$ is irrational using the $\sqrt{2}$ pattern.

2 · Polynomials

Unit: Algebra · 20 marks for chapters 2–5

Basics

- Degree = highest power of x. Linear (1), quadratic (2), cubic (3).
- **Zero** of p(x): a value k with p(k) = 0. On a graph, zeros are the x-coordinates where y = p(x) meets the x-axis.
- A polynomial of degree n has **at most n** zeros.
- Graph of $ax^2 + bx + c$ is a parabola: opens up if $a > 0$, down if $a < 0$. It cuts the x-axis at 2, 1 or 0 points.

Zeros and coefficients — quadratic $ax^2 + bx + c$

$$\alpha + \beta = -\frac{b}{a} \qquad \alpha\beta = \frac{c}{a}$$

Build a polynomial from its zeros:

$$p(x) = k\,[\,x^2 - (\alpha + \beta)x + \alpha\beta\,]$$

Useful rearrangements:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$
$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$
$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

EXTRA Cubic $ax^3 + bx^2 + cx + d$: $\alpha+\beta+\gamma = -\frac{b}{a}$, $\alpha\beta+\beta\gamma+\gamma\alpha = \frac{c}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$.

More vocabulary

- $p(x) = a_n x^n + \ldots + a_1 x + a_0$ with whole-number powers. $\frac{1}{x}$, $\sqrt{x}$ and x^{-2} are not polynomial terms.
- Monomial (1 term), binomial (2), trinomial (3).
- Constant polynomial (e.g. 7) has degree 0. The degree of the zero polynomial is not defined.
- Value of p(x) at x = k is p(k). A linear polynomial ax + b has exactly one zero, $-\frac{b}{a}$.

Reading a graph: number of zeros = number of points where the graph meets the x-axis. A graph that only touches the axis counts once (equal zeros).

Example · **Find the zeros and verify the relationship**

(a) $x^2 - 2x - 8 = (x - 4)(x + 2)$. Zeros: 4 and –2.

$$\text{Sum} = 2 = -\frac{(-2)}{1} \checkmark \qquad \text{Product} = -8 = \frac{-8}{1} \checkmark$$

(b) $6x^2 - 3 - 7x$. First arrange: $6x^2 - 7x - 3 = (3x + 1)(2x - 3)$. Zeros: $-\frac{1}{3}$ and $\frac{3}{2}$.

$$\text{Sum} = \frac{7}{6} = -\frac{(-7)}{6} \checkmark \qquad \text{Product} = -\frac{1}{2} = \frac{-3}{6} \checkmark$$

(c) $x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$. Zeros $\pm\sqrt{3}$; sum 0, product –3.

Example · **Form a polynomial; use α and β**

Sum √2, product $\frac{1}{3}$: $x^2 - \sqrt{2}\,x + \frac{1}{3}$. Multiply by 3 → $3x^2 - 3\sqrt{2}\,x + 1$.

If α, β are zeros of $x^2 - 5x + 6$: α + β = 5, αβ = 6.

$$\alpha^2 + \beta^2 = 25 - 12 = 13 \qquad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{5}{6}$$
$$(\alpha - \beta)^2 = 25 - 24 = 1 \qquad \alpha^3 + \beta^3 = 125 - 90 = 35$$

New polynomial with zeros 2α and 2β: sum 10, product 24 → $x^2 - 10x + 24$.

Quick facts for 1-mark questions

- Zeros are equal in size, opposite in sign ($\alpha = -\beta$) ⇒ sum = 0 ⇒ b = 0.
- Zeros are reciprocals ($\alpha\beta = 1$) ⇒ c = a.
- One zero is 0 ⇒ c = 0.
- Polynomial with zeros $\frac{1}{\alpha}$, $\frac{1}{\beta}$ is $cx^2 + bx + a$ (coefficients reversed).
- If k is a zero, substitute x = k and set p(k) = 0 to find an unknown coefficient.
- Infinitely many polynomials share the same zeros: multiply by any non-zero k.

EXTRA **Division algorithm:** $p(x) = g(x) \times q(x) + r(x)$, where r(x) = 0 or degree of r < degree of g.

3 · Pair of Linear Equations in Two Variables

Consistency test

For $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$:

Compare ratios	Graph	Solutions	Called
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	intersecting	exactly one	consistent
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	coincident	infinitely many	consistent (dependent)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	parallel	none	inconsistent

Watch out: Bring both equations to the form ax + by + c = 0 before comparing ratios.

Solving methods

- **Graphical:** plot both lines (3 points each); the intersection is the solution.
- **Substitution:** from one equation write y in terms of x, put it in the other.
- **Elimination:** multiply to make the coefficients of one variable equal, then add or subtract.

Tip: Always substitute the answer back in both equations to check.

Setting up word problems

- **Two-digit number:** tens digit x, units digit y → number 10x + y; reversed 10y + x.
- **Fraction:** numerator x, denominator y → $\frac{x}{y}$.
- **Ages:** n years ago → (x – n); n years later → (x + n).
- **Boat:** speed in still water x, stream y → downstream x + y, upstream x – y. Time = $\frac{\text{distance}}{\text{speed}}$.
- **Fixed + variable charge:** total = x + (units) × y.
- **Supplementary / cyclic quadrilateral angles:** use the 180° sums.

Example · Elimination, step by step

Solve 2x + 3y = 13 and 5x – 2y = 4.

$$(1) \times 2: \quad 4x + 6y = 26$$
$$(2) \times 3: \quad 15x - 6y = 12$$
$$\text{Add:} \quad 19x = 38 \Rightarrow x = 2$$
$$\text{Put in (1):} \quad 4 + 3y = 13 \Rightarrow y = 3$$

Check in (2): 10 – 6 = 4 ✓

Swapped coefficients (99x + 101y = 499, 101x + 99y = 501): add → x + y = 5; subtract → x – y = 1; so x = 3, y = 2.

Example · Finding k

No solution: kx + 3y = k – 3 and 12x + ky = k.

$$\frac{k}{12} = \frac{3}{k} \Rightarrow k^2 = 36 \Rightarrow k = \pm 6$$

Also need $\frac{3}{k} \neq \frac{k-3}{k}$, i.e. k ≠ 6. So **k = –6**. (k = 6 gives coincident lines.)

Infinitely many solutions: 2x + 3y = 7 and (a – b)x + (a + b)y = 3a + b – 2.

$$\frac{2}{a-b} = \frac{3}{a+b} = \frac{7}{3a+b-2}$$
$$\text{First two: } a = 5b. \text{ Last two: } 2a - 4b = 6.$$
$$\Rightarrow b = 1, a = 5$$

Example · Word problems solved

Digits: the sum of the digits of a two-digit number is 9; nine times the number equals twice the number with digits reversed.

$$x + y = 9 \quad 9(10x + y) = 2(10y + x) \Rightarrow y = 8x$$
$$x = 1, y = 8 \Rightarrow \text{number} = 18$$

Boat: 30 km upstream + 44 km downstream takes 10 h; 40 km upstream + 55 km downstream takes 13 h. Let u = $\frac{1}{x-y}$, v = $\frac{1}{x+y}$.

$$30u + 44v = 10 \quad 40u + 55v = 13$$
$$\Rightarrow u = \frac{1}{5}, v = \frac{1}{11} \Rightarrow x - y = 5, x + y = 11$$
$$\text{Boat } 8 \text{ km/h, stream } 3 \text{ km/h}$$

Graph questions

- Make a table of 3 points for each line, plot on the same axes, read the intersection.
- **Triangle with the x-axis:** vertices are the intersection point and the two x-intercepts. Area = ½ × base on the axis × height (the y-coordinate of the intersection).
- Line through the origin: c = 0. Lines x = a and y = b meet at (a, b).

EXTRA **Reducible equations:** for $\frac{2}{x} + \frac{3}{y} = 13$, put $\frac{1}{x} = u$ and $\frac{1}{y} = v$, solve the linear pair, then invert.

4 · Quadratic Equations

Standard form and solving

$$ax^2 + bx + c = 0, \quad a \neq 0$$

1. Factorisation (splitting the middle term): find two numbers whose *sum is b* and *product is ac*. Example: $2x^2 - 5x + 3$ → sum −5, product 6 → −2 and −3 → $(2x - 3)(x - 1) = 0 \rightarrow x = \frac{3}{2}, 1$.

2. Quadratic formula:

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

Discriminant and nature of roots

$$D = b^2 - 4ac$$

D	Roots
$D > 0$	two distinct real roots
$D = 0$	two equal real roots, each $-\frac{b}{2a}$
$D < 0$	no real roots

"Real roots" means $D \geq 0$. "Equal roots — find k" means set $D = 0$.

Sum of roots $= -\frac{b}{a}$, product $= \frac{c}{a}$.

Word problems

- Consecutive integers: x, x + 1. Consecutive odd or even: x, x + 2.
- Speed problems: $\frac{d}{x} - \frac{d}{x+k}$ = time difference → take LCM → quadratic.
- Rectangle: length × breadth = area with one side written in terms of the other.
- Right triangle: use Pythagoras to form the equation.

Watch out: Reject roots that make no sense (negative length, speed, age). Write "$x \neq -5$ since speed cannot be negative".

Is it quadratic? Simplify first

- $(x + 1)^2 = 2(x - 3) \Rightarrow x^2 + 7 = 0$. Quadratic.
- $x(x + 1) + 8 = (x + 2)(x - 2) \Rightarrow x + 12 = 0$. **Not** quadratic — x^2 cancels.
- $x + \frac{1}{x} = 3 \Rightarrow x^2 - 3x + 1 = 0$ (multiply by x). Quadratic.

Root given, find k: substitute the root. If $x = 2$ is a root of $x^2 + kx - 10 = 0$, then $4 + 2k - 10 = 0 \Rightarrow k = 3$.

Roots given, form the equation: x^2 − (sum of roots)x + (product of roots) = 0.

Example · Formula and discriminant

Solve $2x^2 - 7x + 3 = 0$. a = 2, b = −7, c = 3.

$$D = 49 - 24 = 25 \quad x = \frac{7 \pm 5}{4} = 3 \text{ or } \frac{1}{2}$$

Equal roots: $kx(x - 2) + 6 = 0 \Rightarrow kx^2 - 2kx + 6 = 0$.

$$D = 4k^2 - 24k = 0 \Rightarrow 4k(k - 6) = 0 \Rightarrow k = 6$$

k = 0 is rejected because the equation would stop being quadratic.

$2x^2 + kx + 3 = 0$, equal roots: $k^2 = 24 \Rightarrow k = \pm 2\sqrt{6}$.

Example · Word problems solved

Train: 360 km at uniform speed; 5 km/h faster would take 1 hour less.

$$\frac{360}{x} - \frac{360}{x+5} = 1 \Rightarrow x^2 + 5x - 1800 = 0$$
$$(x + 45)(x - 40) = 0 \Rightarrow x = 40 \text{ km/h}$$

Consecutive integers: the sum of squares of two consecutive positive integers is 365.

$$x^2 + (x + 1)^2 = 365 \Rightarrow x^2 + x - 182 = 0$$
$$(x + 14)(x - 13) = 0 \Rightarrow 13 \text{ and } 14$$

Taps: two taps together fill a tank in $\frac{75}{8}$ hours; the larger takes 10 hours less than the smaller.

$$\frac{1}{x} + \frac{1}{x-10} = \frac{8}{75} \Rightarrow 8x^2 - 230x + 750 = 0$$
$$\Rightarrow x = 25 \text{ (smaller tap), larger tap 15 hours}$$

EXTRA Completing the square

1. Make the coefficient of x^2 equal to 1.
2. Move the constant to the right.
3. Add (half the coefficient of x)^2 to both sides.
4. Write the left side as a perfect square and take the square root.

$$x^2 + 6x - 7 = 0 \Rightarrow x^2 + 6x + 9 = 16$$
$$(x + 3)^2 = 16 \Rightarrow x + 3 = \pm 4 \Rightarrow x = 1 \text{ or } -7$$

The quadratic formula is this method applied to $ax^2 + bx + c = 0$.

5 · Arithmetic Progressions

Core formulas

AP: a, a + d, a + 2d, … where a = first term, d = common difference = $a_2 - a_1$.

nth term: $a_n = a + (n-1)d$

Sum: $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$

l = last term. Use the second form whenever the last term is known.

$a_n = S_n - S_{n-1}$ nth term from the end $= l - (n-1)d$

Shortcuts

- a, b, c are in AP ⇔ 2b = a + c.
- Choose 3 terms as a − d, a, a + d; 4 terms as a − 3d, a − d, a + d, a + 3d (sum problems become easy).
- Number of terms: $n = \frac{l - a}{d} + 1$.
- Sum of first n natural numbers = $\frac{n(n+1)}{2}$.
- Sum of first n odd numbers = n^2. Sum of first n even numbers = n(n + 1).
- If a_n is of the form pn + q, the sequence is an AP with d = p.

Watch out: n must be a positive integer. If you get n = 12.5, the term is not in the AP.

Example · nth term problems

10th term of 2, 7, 12 … a = 2, d = 5 ⇒ $a_{10} = 2 + 9 \times 5 = 47$.

Which term of 21, 18, 15 … is –81?

$21 + (n-1)(-3) = -81 \Rightarrow n - 1 = 34 \Rightarrow n = 35$

Given $a_3 = 5$ and $a_7 = 9$: subtract → 4d = 4 ⇒ d = 1, a = 3. AP: 3, 4, 5, 6 …

How many multiples of 4 lie between 10 and 250? 12, 16 … 248.

$n = \frac{248 - 12}{4} + 1 = 60$

Three-digit numbers divisible by 7: 105 … 994 ⇒ n = 128.

Example · Sum problems

34 + 32 + 30 + … + 10: $n = \frac{10 - 34}{-2} + 1 = 13$.

$S = \frac{13}{2}(34 + 10) = 286$

How many terms of 24, 21, 18 … give a sum of 78?

$\frac{n}{2}[48 - 3(n-1)] = 78 \Rightarrow n^2 - 17n + 52 = 0$

$\Rightarrow n = 4 \text{ or } 13$

Both are valid: terms 5 to 13 add up to zero.

Given $S_n = 4n - n^2$: $a_1 = S_1 = 3$; $S_2 = 4$ so $a_2 = 1$; d = –2; $a_n = 5 - 2n$.

Example · Real-life AP

200 logs are stacked with 20 in the bottom row, 19 in the next, 18 in the next … How many rows?

$\frac{n}{2}[40 - (n-1)] = 200 \Rightarrow n^2 - 41n + 400 = 0$

$\Rightarrow n = 16 \text{ or } 25$

n = 25 would need a negative number of logs in the top row, so n = 16 rows and the top row has 20 − 15 = 5 logs.

Other AP situations: fixed yearly salary increment, equal monthly increase in savings, taxi fare per km after a fixed charge, rungs of a ladder. Compound interest is **not** an AP.

Properties for tricky questions

- Adding, subtracting, multiplying or dividing every term by the same non-zero number still gives an AP.
- Sum of terms equidistant from the two ends = a + l.
- If $S_n = An^2 + Bn$, it is an AP with d = 2A.
- If $a_m = n$ and $a_n = m$, then d = –1 and $a_{m+n} = 0$.
- If $m \times a_m = n \times a_n$, then $a_{m+n} = 0$.
- **Ratio of sums → ratio of terms:** replace n by (2m − 1) in the ratio of S_n to get the ratio of the mth terms.
- $a_n - a_k = (n - k)d$ — the fastest way to find d from two terms.

6 · Triangles

Unit: Geometry · 15 marks with Circles

Similarity

Two triangles are similar (~) if corresponding angles are equal **and** corresponding sides are in the same ratio. Congruent = same shape and size; similar = same shape only.

$$\Delta ABC \sim \Delta PQR \Rightarrow \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

Criteria:

- **AA (AAA):** two pairs of angles equal.
- **SSS:** all three pairs of sides proportional.
- **SAS:** one angle equal and the sides including it proportional.

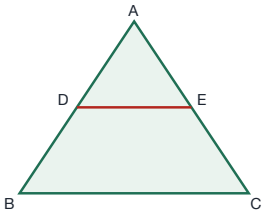
Watch out: The order of letters matters. Write the correspondence correctly (A↔P, B↔Q, C↔R) before writing ratios.

Basic Proportionality Theorem (Thales)

If DE || BC in ΔABC, then

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Also $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$.



Converse: if a line divides two sides in the same ratio, it is parallel to the third side.

The proof of BPT (using areas of triangles on the same base) is a frequently asked 5-mark question.

Results that follow from similarity

- Ratio of corresponding sides = ratio of perimeters = ratio of corresponding altitudes = ratio of corresponding medians.
- EXTRA Ratio of areas = (ratio of corresponding sides)².
- In a right triangle with altitude to the hypotenuse, all three triangles formed are similar.
- Shadow problems: object and shadow form similar triangles → $\frac{\text{height}_1}{\text{shadow}_1} = \frac{\text{height}_2}{\text{shadow}_2}$.

Proof · Basic Proportionality Theorem

Given: ΔABC with DE || BC, D on AB, E on AC. **To prove:** $\frac{AD}{DB} = \frac{AE}{EC}$.

Construction: join BE and CD. Draw EN ⊥ AB and DM ⊥ AC.

1. ar(ADE) = ½ × AD × EN and ar(BDE) = ½ × DB × EN, so $\frac{ar(ADE)}{ar(BDE)} = \frac{AD}{DB}$.
2. ar(ADE) = ½ × AE × DM and ar(DEC) = ½ × EC × DM, so $\frac{ar(ADE)}{ar(DEC)} = \frac{AE}{EC}$.
3. ΔBDE and ΔDEC are on the same base DE and between the same parallels DE and BC, so ar(BDE) = ar(DEC).
4. From 1, 2 and 3: $\frac{AD}{DB} = \frac{AE}{EC}$.

Example · Using BPT and similarity

BPT: DE || BC, AD = 1.5 cm, DB = 3 cm, AE = 1 cm. Then $\frac{1.5}{3} = \frac{1}{EC} \Rightarrow EC = 2$ cm.

Converse: if $\frac{AD}{DB} = \frac{AE}{EC}$, state "by the converse of BPT, DE || BC".

Shadow: a 6 m pole casts a 4 m shadow when a tower casts a 28 m shadow.

$$\frac{h}{28} = \frac{6}{4} \Rightarrow h = 42 \text{ m}$$

Lamp post: a girl 90 cm tall walks from a 3.6 m lamp post at 1.2 m/s. After 4 s she is 4.8 m away; let her shadow be x.

$$\frac{x}{4.8 + x} = \frac{0.9}{3.6} \Rightarrow 4x = 4.8 + x \Rightarrow x = 1.6 \text{ m}$$

How to spot similar triangles

- A **common angle** shared by both triangles.
- **Vertically opposite angles** where two lines cross.
- **Parallel lines** → alternate or corresponding angles equal.
- Two **right angles**, plus one more equal angle.

Two equal angles are enough (AA). Then write the similarity in the correct order and pick the ratio you need.

Trapezium ABCD (AB || DC), diagonals meet at O: ΔAOB ~ ΔCOD by AA, so $\frac{AO}{OC} = \frac{BO}{OD}$.

EXTRA **Angle bisector theorem:** if AD bisects ∠A, then $\frac{BD}{DC} = \frac{AB}{AC}$.

Proof · Pythagoras theorem by similarity

Given: ΔABC right-angled at B. **To prove:** $AC^2 = AB^2 + BC^2$. **Construction:** BD ⊥ AC.

1. ΔADB ~ ΔABC (∠A common, right angles) ⇒ $\frac{AD}{AB} = \frac{AB}{AC} \Rightarrow AB^2 = AD \times AC$.
2. ΔBDC ~ ΔABC (∠C common, right angles) ⇒ $BC^2 = CD \times AC$.
3. Add: $AB^2 + BC^2 = AC(AD + CD) = AC^2$.

The proof is outside the current syllabus EXTRA, but the theorem is used everywhere. Related results: in an equilateral triangle of side a and height h, $4h^2 = 3a^2$. In a rhombus, $4 \times \text{side}^2 = d_1^2 + d_2^2$.

7 · Coordinate Geometry 6 marks

Formulas

$$\text{Distance AB} = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

Distance of (x, y) from the origin = $\sqrt{(x^2 + y^2)}$.

Section formula (m : n, internal):

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

$$\text{Mid-point} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

EXTRA Centroid = $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$. Area of triangle = $\frac{1}{2} \left| x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2) \right|$.

How to use them

- Point on the x-axis is (x, 0); on the y-axis is (0, y).
- **Finding a ratio:** assume k : 1, apply the section formula, solve for k.
- **Collinear points:** AB + BC = AC (largest distance = sum of the other two).
- **Equidistant point:** set $PA^2 = PB^2$ (squaring removes the roots).
- Signs by quadrant: I (+,+) · II (–,+) · III (–,–) · IV (+,–).

To prove	Show
Parallelogram	mid-points of both diagonals coincide
Rectangle	opposite sides equal + diagonals equal
Rhombus	all four sides equal
Square	all four sides equal + diagonals equal
Right triangle	largest side² = sum of the other two squares

Example · Distance formula

Distance between (2, 3) and (4, 1): $\sqrt{(2^2 + (-2)^2)} = \sqrt{8} = 2\sqrt{2}$.

Point on the x-axis equidistant from (2, –5) and (–2, 9): let it be (x, 0).

$$(x-2)^2 + 25 = (x+2)^2 + 81$$
$$-4x + 29 = 4x + 85 \Rightarrow x = -7 \Rightarrow (-7, 0)$$

Type of triangle (5, –2), (6, 4), (7, –2): AB = $\sqrt{37}$, BC = $\sqrt{37}$, AC = 2. Two sides equal ⇒ isosceles.

Distance of (x, y) from the x-axis is |y|; from the y-axis is |x|.

Example · Section formula

Point dividing (4, –3) and (8, 5) in the ratio 3 : 1:

$$\left(\frac{3\times 8 + 1\times 4}{4}, \frac{3\times 5 + 1\times(-3)}{4}\right) = (7, 3)$$

Ratio in which the y-axis divides (5, –6) and (–1, –4): let it be k : 1. The x-coordinate is 0.

$$\frac{-k+5}{k+1} = 0 \Rightarrow k = 5 \Rightarrow \text{ratio } 5:1$$
$$y = \frac{-20-6}{6} = -\frac{13}{3} \Rightarrow \text{point } \left(0, -\frac{13}{3}\right)$$

Trisection of (4, –1) and (–2, –3): use ratios 1 : 2 and 2 : 1 → $\left(2, -\frac{5}{3}\right)$ and $\left(0, -\frac{7}{3}\right)$.

Example · Parallelogram and circle

(1, 2), (4, y), (x, 6), (3, 5) are vertices of a parallelogram in order. Diagonals bisect each other, so mid-points are equal.

$$\frac{1+x}{2} = \frac{4+3}{2} \Rightarrow x = 6 \qquad \frac{2+6}{2} = \frac{y+5}{2} \Rightarrow y = 3$$

Shortcut for the fourth vertex: D = A + C – B (coordinate-wise).

Circle: the centre is the mid-point of a diameter; radius = distance from the centre to any point on the circle. If one end of a diameter is (2, 3) and the centre is (–2, 5), the other end is (2 × (–2) – 2, 2 × 5 – 3) = (–6, 7).

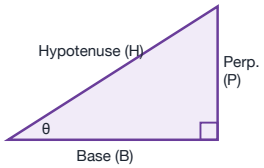
Shortcuts

- x-axis divides the join of (x_1, y_1) and (x_2, y_2) in the ratio $-y_1 : y_2$.
- y-axis divides it in the ratio $-x_1 : x_2$.
- A negative ratio means the point divides the segment externally.
- Median from A goes to the mid-point of BC. The centroid divides each median 2 : 1 from the vertex.
- Work with squared distances for as long as possible to avoid square roots.
- EXTRA Three points are collinear if the triangle area formula gives 0.

8 · Introduction to Trigonometry

Unit: Trigonometry · 12 marks with chapter 9

The six ratios



$$\sin\theta = \frac{P}{H} \quad \cos\theta = \frac{B}{H} \quad \tan\theta = \frac{P}{B}$$

$$\text{cosec}\,\theta = \frac{H}{P} \quad \sec\theta = \frac{H}{B} \quad \cot\theta = \frac{B}{P}$$

P = side *opposite* θ, B = side *adjacent* to θ.

$$\text{cosec}\,\theta = \frac{1}{\sin\theta} \quad \sec\theta = \frac{1}{\cos\theta} \quad \cot\theta = \frac{1}{\tan\theta}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

Tip: Given one ratio, draw a right triangle, fill two sides, get the third by Pythagoras, then read off all other ratios.

Standard values

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	n.d.
cosec	n.d.	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	n.d.
cot	n.d.	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

n.d. = not defined. **Memory trick for sin:** √(0/4), √(1/4), √(2/4), √(3/4), √(4/4). cos is the same row reversed.

sin θ and cos θ always lie between 0 and 1 for acute θ; sec θ and cosec θ are always ≥ 1.

Identities

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \text{cosec}^2\theta$$

Rearranged: $\sec^2\theta - \tan^2\theta = 1 \Rightarrow (\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$. Same for cosec and cot.

EXTRA **Complementary angles:** $\sin(90° - \theta) = \cos\theta \cdot \tan(90° - \theta) = \cot\theta \cdot \sec(90° - \theta) = \text{cosec}\,\theta$.

Watch out: $\sin^2\theta$ means $(\sin\theta)^2$. $\sin(A + B) \neq \sin A + \sin B$.

How to prove an identity

- Start from the more complicated side (usually LHS).
- Convert everything to sin and cos if you are stuck.
- Take LCM when adding fractions.
- Multiply by the conjugate when you see $(1 \pm \sin\theta)$ or $(\sec\theta \pm \tan\theta)$ in a denominator.
- Replace 1 by $\sin^2\theta + \cos^2\theta$ (or $\sec^2\theta - \tan^2\theta$) when needed.
- Use $a^2 - b^2$, $(a \pm b)^2$ and $a^3 \pm b^3$ identities.

Example · All ratios from one ratio

Given sin A = $\frac{3}{4}$: P = 3, H = 4, so B = √(16 − 9) = √7.

$$\cos A = \frac{\sqrt{7}}{4} \quad \tan A = \frac{3}{\sqrt{7}} \quad \sec A = \frac{4}{\sqrt{7}}$$

Given 15 cot A = 8: $\cot A = \frac{8}{15}$, so B = 8, P = 15, H = 17.

$$\sin A = \frac{15}{17} \quad \sec A = \frac{17}{8}$$

The ratios depend only on the angle, not on the size of the triangle.

Example · Evaluating with standard values

$$\sin 60° \cos 30° + \sin 30° \cos 60° = \frac{3}{4} + \frac{1}{4} = 1$$

$$2\tan^2 45° + \cos^2 30° - \sin^2 60° = 2 + \frac{3}{4} - \frac{3}{4} = 2$$

$$\frac{5\cos^2 60° + 4\sec^2 30° - \tan^2 45°}{\sin^2 30° + \cos^2 30°} = \frac{5}{4} + \frac{16}{3} - 1 = \frac{67}{12}$$

Find A and B: tan(A + B) = √3 and $\tan(A - B) = \frac{1}{\sqrt{3}} \Rightarrow$ A + B = 60°, A − B = 30° ⇒ A = 45°, B = 15°.

Example · Identity proofs worked out

1. $\frac{1 + \tan^2 A}{1 + \cot^2 A} = \frac{\sec^2 A}{\text{cosec}^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A$.

2. $\sqrt{\left(\frac{1 + \sin A}{1 - \sin A}\right)}$: multiply top and bottom inside the root by (1 + sin A).

$$= \sqrt{\left(\frac{(1 + \sin A)^2}{\cos^2 A}\right)} = \frac{1 + \sin A}{\cos A} = \sec A + \tan A$$

3. $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \frac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)} = \tan\theta$, because $1 - 2\sin^2\theta = 2\cos^2\theta - 1$.

4. $(\text{cosec}\,\theta - \cot\theta)^2 = \frac{(1 - \cos\theta)^2}{\sin^2\theta} = \frac{(1 - \cos\theta)^2}{(1 - \cos\theta)(1 + \cos\theta)} = \frac{1 - \cos\theta}{1 + \cos\theta}$.

Handy derived results

$$(\sin\theta+\cos\theta)^2 = 1+2\sin\theta\cos\theta$$
$$(\sin\theta-\cos\theta)^2 = 1-2\sin\theta\cos\theta$$
$$\tan\theta+\cot\theta = \frac{1}{\sin\theta\cos\theta} = \sec\theta\,\operatorname{cosec}\theta$$
$$\sin^4\theta-\cos^4\theta = \sin^2\theta-\cos^2\theta$$
$$\sec^4\theta-\tan^4\theta = \sec^2\theta+\tan^2\theta$$

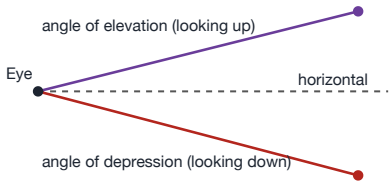
If $\sec\theta+\tan\theta = p$, then $\sec\theta-\tan\theta = \frac{1}{p}$. Add the two to get $\sec\theta$; subtract to get $\tan\theta$.

In terms of sin A: $\cos A = \sqrt{(1-\sin^2 A)}$, $\tan A = \frac{\sin A}{\sqrt{(1-\sin^2 A)}}$.

- As θ goes from 0° to 90°, sin θ rises from 0 to 1 and cos θ falls from 1 to 0. They are equal at 45°.
- "sin A" is one symbol, not sin × A. You cannot cancel the A in $\frac{\sin A}{\cos A}$.

9 · Some Applications of Trigonometry — Heights and Distances

Terms



- **Line of sight:** line from the eye to the object.
- **Angle of elevation:** angle above the horizontal when looking up.
- **Angle of depression:** angle below the horizontal when looking down.
- Angle of depression from A to B = angle of elevation from B to A (alternate angles).

Method

- Draw a neat labelled figure — it carries marks.
- Mark the right angle, the known side and the unknown side.
- Almost always use **$\tan\theta = \frac{\text{height}}{\text{horizontal distance}}$**.
- Two angles → two right triangles → two equations; eliminate the common unknown.
- Add the observer's height at the end if the eye is above the ground.

Angle	height : distance	Triangle sides
30°	h = d / √3	30-60-90 → 1 : √3 : 2
60°	h = d√3	
45°	h = d	45-45-90 → 1 : 1 : √2

Use √3 = 1.732 only if the question asks; otherwise leave the answer with √3 and rationalise.

Example · One triangle, and two angles on the same side

(a) From a point 30 m from the foot of a tower, the angle of elevation of the top is 30°.

$$\tan 30^\circ = \frac{h}{30} \Rightarrow h = \frac{30}{\sqrt{3}} = 10\sqrt{3} \approx 17.32\text{ m}$$

(b) The elevation of the top of a tower is 30°. After walking 20 m towards it, the elevation is 60°. Let the remaining distance be x.

$$h = x\sqrt{3} \quad\text{and}\quad h = \frac{x+20}{\sqrt{3}}$$
$$3x = x+20 \Rightarrow x = 10 \Rightarrow h = 10\sqrt{3} \approx 17.32\text{ m}$$

Example · Elevation and depression together

(a) From the top of a 7 m building, the elevation of the top of a tower is 60° and the depression of its foot is 45°.

$$\text{Depression } 45^\circ \Rightarrow \text{horizontal distance} = 7\text{ m}$$
$$\text{Extra height} = 7\tan 60^\circ = 7\sqrt{3}$$
$$\text{Tower} = 7(\sqrt{3}+1) \approx 19.12\text{ m}$$

(b) A 1.6 m statue stands on a pedestal. From a point on the ground the elevation of the top of the statue is 60° and of the top of the pedestal is 45°.

$$h = d \quad\text{and}\quad h + 1.6 = d\sqrt{3}$$
$$h = \frac{1.6}{\sqrt{3}-1} = 0.8(\sqrt{3}+1) \approx 2.19\text{ m}$$

Common set-ups

- **Shadow and sun's altitude:** shadow = $\frac{\text{height}}{\tan\theta}$. If the shadow is 40 m longer at 30° than at 60°, then $h\sqrt{3} - \frac{h}{\sqrt{3}} = 40 \Rightarrow h = 20\sqrt{3}$ m.
- **Objects on opposite sides** of a tower: add the two horizontal distances. On the same side: subtract.
- **Kite or ladder:** the string or ladder is the hypotenuse — use $\sin\theta = \frac{\text{height}}{\text{length}}$.
- **River width, two poles of equal height:** two right triangles sharing the total width.
- **Moving car, boat, plane or balloon:** distance moved = difference of the two horizontal distances; speed = $\frac{\text{distance}}{\text{time}}$.
- **Observer has height:** work with (object height − eye height), add the eye height back at the end.

Rationalise: $\frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2}$. $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$.

10 · Circles

Tangent basics

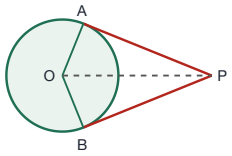
- **Tangent:** touches the circle at exactly one point. **Secant:** cuts it at two points.
- Tangents from a point: inside the circle 0 · on the circle 1 · outside the circle 2.

Theorem 1: the tangent at any point is **perpendicular to the radius** through the point of contact.

Theorem 2: the lengths of the two tangents drawn from an external point are **equal** (PA = PB).

Both proofs are asked in boards. Theorem 2 uses RHS congruence of ΔOAP and ΔOBP.

Results used in problems



$$PA = PB = \sqrt{(OP^2 - r^2)}$$

$$\angle APB + \angle AOB = 180°$$

- OP bisects ∠APB and ∠AOB; OP is the perpendicular bisector of chord AB.
- Quadrilateral ABCD circumscribing a circle: AB + CD = AD + BC.
- Tangents at the two ends of a diameter are parallel.
- A parallelogram circumscribing a circle is a rhombus.
- In-radius of a right triangle with legs a, b and hypotenuse c: $r = \frac{a+b-c}{2}$.

Proof · Tangent is perpendicular to the radius

Given: circle with centre O, tangent XY at P. **To prove:** OP ⊥ XY.

1. Take any point Q on XY other than P and join OQ.
2. Q must lie outside the circle; otherwise XY would cut the circle at two points and be a secant.
3. So OQ > radius, i.e. OQ > OP, for every such Q.
4. OP is therefore the shortest distance from O to the line XY.
5. The shortest distance from a point to a line is the perpendicular. Hence OP ⊥ XY.

Proof · Tangents from an external point are equal

Given: tangents PA and PB from external point P to a circle with centre O. **To prove:** PA = PB.

Construction: join OA, OB, OP.

1. ∠OAP = ∠OBP = 90° (tangent ⊥ radius).
2. OA = OB (radii).
3. OP = OP (common hypotenuse).
4. ΔOAP ≅ ΔOBP (RHS).
5. PA = PB (CPCT). Also ∠APO = ∠BPO, so OP bisects ∠APB.

Example · Standard problems

- **Tangent length:** a point is 5 cm from the centre and the tangent is 4 cm long ⇒ r = √(25 − 16) = 3 cm.
- **Concentric circles** of radii 5 cm and 3 cm: a chord of the larger circle touching the smaller has length 2√(25 − 9) = 8 cm. The point of contact bisects the chord.
- **Angle between tangents is 80°:** ∠AOB = 100°, so ∠POA = 50°.
- **Tangents PT, PQ and chord TQ:** ∠TPQ = 2∠OTQ.

Triangle around a circle: ΔABC circumscribes a circle of radius 4 cm; the circle touches BC at D with BD = 8 cm, DC = 6 cm. Let the tangent length from A be x.

$$AB = x + 8,\ AC = x + 6,\ BC = 14,\ s = x + 14$$
$$\sqrt{((x + 14) \times x \times 8 \times 6)} = 4(x + 14)$$
$$3x = x + 14 \Rightarrow x = 7 \Rightarrow AB = 15\text{ cm},\ AC = 13\text{ cm}$$

More results

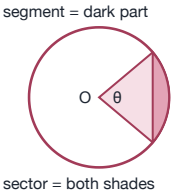
- Area of a triangle = r × s, where r is the in-radius and s the semi-perimeter.
- If the tangent lengths from the three vertices are x, y, z, the sides are x + y, y + z, z + x and s = x + y + z.
- Opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre.
- The perpendicular at the point of contact to the tangent passes through the centre.
- There is exactly one tangent at any point on a circle.
- EXTRA **Alternate segment theorem:** the angle between a tangent and a chord equals the angle in the alternate segment.

Tip: Whenever you see a tangent, join the centre to the point of contact and mark 90°. Most problems then reduce to Pythagoras or angle sum.

11 · Areas Related to Circles

Unit: Mensuration · 10 marks with chapter 12

Sector and segment



$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r$$

$$\text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Segment area} = \text{sector area} - \text{triangle area}$$

Triangle area: for θ = 90° it is ½r²; for θ = 60° it is $\frac{\sqrt{3}}{4}r^2$ (equilateral); for θ = 120° it is $\frac{\sqrt{3}}{4}r^2$. In general ½ r² sin θ.

Major sector or segment = πr² − minor one. Sector area also = ½ × arc length × r.

Other formulas

- Circumference 2πr = πd. Area πr².
- Perimeter of a sector = 2r + arc length.
- Semicircle: area ½πr², perimeter πr + 2r. Quadrant: area ¼πr².
- Ring (between two concentric circles) = π(R² − r²).
- Distance covered by a wheel in one revolution = its circumference. Revolutions = $\frac{\text{distance}}{2\pi r}$.
- **Clock:** minute hand sweeps 6° per minute; hour hand 30° per hour (0.5° per minute).

Tip: Use π = 22/7 unless the question says 3.14. If r is a multiple of 7, 22/7 cancels neatly.

Example · Sector and segment

Sector, r = 4 cm, θ = 30° (π = 3.14):

$$\text{Area} = \frac{30}{360} \times 3.14 \times 16 \approx 4.19\text{ cm}^2$$
$$\text{Major sector} = 50.24 - 4.19 \approx 46.05\text{ cm}^2$$

Chord subtends 90°, r = 10 cm:

$$\text{Minor sector} = \tfrac{1}{4} \times 3.14 \times 100 = 78.5\text{ cm}^2$$
$$\text{Triangle} = \tfrac{1}{2} \times 10 \times 10 = 50\text{ cm}^2$$
$$\text{Minor segment} = 28.5\text{ cm}^2 \quad \text{Major sector} = 235.5\text{ cm}^2$$

Segment, r = 21 cm, θ = 120°:

$$\text{Sector} = \frac{1}{3} \times \frac{22}{7} \times 441 = 462\text{ cm}^2$$
$$\text{Triangle} = \frac{\sqrt{3}}{4} \times 441\text{ cm}^2$$
$$\text{Segment} = 462 - \frac{441\sqrt{3}}{4} = \frac{21}{4}(88 - 21\sqrt{3})\text{ cm}^2$$

Example · Clock, wheel, arc

Clock: minute hand 14 cm; area swept in 5 minutes. Angle = 5 × 6° = 30°.

$$\frac{30}{360} \times \frac{22}{7} \times 196 = \frac{154}{3} \approx 51.33\text{ cm}^2$$

Wheel: diameter 80 cm, car speed 66 km/h. Revolutions in 10 minutes?

$$\text{Distance} = 11\text{ km} = 1\,100\,000\text{ cm}$$
$$\text{Circumference} = \frac{22}{7} \times 80\text{ cm}$$
$$\text{Revolutions} = 1\,100\,000 \times \frac{7}{1760} = 4375$$

Arc: r = 21 cm, θ = 60° ⇒ arc = $\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm.

Triangle inside the sector

θ	Triangle area	Chord length
60°	$\frac{\sqrt{3}}{4}r^2$ (equilateral)	r
90°	½ r^2	$r\sqrt{2}$
120°	$\frac{\sqrt{3}}{4}r^2$	$r\sqrt{3}$

For 120°: drop a perpendicular from the centre to the chord. It bisects the angle (60° each), so height = r cos 60° = $\frac{r}{2}$ and half-chord = r sin 60° = $\frac{\sqrt{3}}{2}$ r.

Shaded-region strategies

- Shaded area = area of the outer figure − area of the unshaded parts.
- **Circle inside a square:** diameter = side. **Square inside a circle:** diagonal = diameter.
- **Four quadrants at the corners of a square** = one full circle.
- **Sectors at the three vertices of an equilateral triangle** (60° each) = half a circle in total.
- **Animal tied at a corner** of a square or rectangular field grazes a quadrant; at a corner of a triangular field, a sector with that corner's angle.
- **Umbrella with 8 ribs, brooch, table design:** equal sectors of $\frac{360°}{n}$ each.
- **Wiper blades:** each sweeps a sector; add if they do not overlap.
- **Track with semicircular ends:** two straight parts + one full circle.

12 · Surface Areas and Volumes

Combination of solids

Volume of a combined solid = sum of the individual volumes.

Surface area = sum of only the **visible** surfaces — never add full TSAs, because the joined faces disappear.

Shape	Surface area
Cone on hemisphere (toy, top)	$\pi rl + 2\pi r^2$
Cylinder + cone (tent)	$2\pi rh + \pi rl$
Cylinder + 2 hemispheres (capsule)	$2\pi rh + 4\pi r^2$
Hemisphere scooped from a cube	$6a^2 - \pi r^2 + 2\pi r^2$
Cylinder with cone hollowed out	$2\pi rh + \pi r^2 + \pi rl$

Formulas for the individual solids are in the Part A table.

Conversion and flow problems

- **Melting and recasting:** volume stays the same. Number of small objects = $\frac{\text{volume of big solid}}{\text{volume of one small solid}}$.
- **Water flowing through a pipe:** volume in time t = area of cross-section × speed × t.
- **Solid dropped in water:** volume of water displaced (or rise × base area) = volume of the solid.
- Keep all lengths in the same unit before calculating. Diameter given? Halve it first.

EXTRA **Frustum of a cone:** $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$, $CSA = \pi(R + r)l$, $l = \sqrt{(h^2 + (R - r)^2)}$.

Example · Surface area of combined solids

Toy: a cone on a hemisphere, radius 3.5 cm, total height 15.5 cm.

$$\text{Cone height} = 15.5 - 3.5 = 12 \text{ cm}$$
$$l = \sqrt{(3.5^2 + 12^2)} = 12.5 \text{ cm}$$
$$\text{TSA} = \pi rl + 2\pi r^2 = \pi r(l + 2r)$$
$$= \frac{22}{7} \times 3.5 \times 19.5 = 214.5 \text{ cm}^2$$

Capsule: cylinder with two hemispherical ends, total length 14 mm, diameter 5 mm.

$$\text{Cylinder length} = 14 - 5 = 9 \text{ mm}$$
$$\text{Area} = 2\pi rh + 4\pi r^2 = 2\pi r(h + 2r)$$
$$= 2 \times \frac{22}{7} \times 2.5 \times 14 = 220 \text{ mm}^2$$

Example · Recasting and flow

Melting: a metal sphere of radius 4.2 cm is recast into a cylinder of radius 6 cm.

$$\frac{4}{3}\pi (4.2)^3 = \pi \times 6^2 \times h$$
$$h = \frac{4 \times 74.088}{3 \times 36} = 2.744 \text{ cm}$$

Canal: 6 m wide, 1.5 m deep, water flowing at 10 km/h. Area irrigated in 30 minutes if 8 cm of standing water is needed?

$$\text{Length of water in 30 min} = 5000 \text{ m}$$
$$\text{Volume} = 6 \times 1.5 \times 5000 = 45\,000 \text{ m}^3$$
$$\text{Area} = \frac{45\,000}{0.08} = 562\,500 \text{ m}^2 = 56.25 \text{ hectares}$$

Counting: a sphere of radius 3 cm makes $\frac{3^3}{1^3} = 27$ balls of radius 1 cm.

More formulas and facts

- **Hollow cylinder (pipe):** volume = $\pi(R^2 - r^2)h$; curved surface (inner + outer) = $2\pi(R + r)h$.
- Cone : hemisphere : cylinder with the same radius and height = 1 : 2 : 3 in volume.
- A cone has one-third the volume of a cylinder with the same base and height.
- **Scaling:** if every length is multiplied by k, surface area × k^2, volume × k^3. Doubling the radius of a sphere gives 4 times the area and 8 times the volume.
- **Largest sphere in a cube** of edge a: diameter = a. **Largest cone in a cube:** $r = \frac{a}{2}$, h = a.
- **Cube inside a sphere:** $a\sqrt{3} = 2r$.

Other standard situations

- **Well and embankment:** earth dug out $\pi r^2 h$ = volume of the ring $\pi(R^2 - r^2)H$ spread around it.
- **Wall and bricks:** number of bricks = $\frac{\text{volume of wall}}{\text{volume of one brick}}$ (subtract mortar if given).
- **Rain on a roof:** roof area × rainfall depth = volume collected in the tank.
- **Ice-cream cone:** cone + hemisphere; number of cones = $\frac{\text{volume of container}}{\text{volume of one filled cone}}$.
- **Vessel = hemisphere + cylinder:** inner surface = $2\pi r^2 + 2\pi rh$.
- **Wooden article with scoops:** the scooped surface is still exposed — add the curved area of each scoop, subtract the flat circles removed.
- **Cost questions:** cost = rate × area (painting, canvas) or rate × volume (milk, digging).

13 · Statistics

Unit: Statistics and Probability · 11 marks

Mean of grouped data

Class mark $x_i = \frac{\text{upper limit} + \text{lower limit}}{2}$; f_i = frequency.

$$\text{Direct: } \bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$$

$$\text{Assumed mean: } \bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}, \quad d_i = x_i - a$$

$$\text{Step deviation: } \bar{x} = a + \left(\frac{\Sigma f_i u_i}{\Sigma f_i}\right) \times h, \quad u_i = \frac{x_i - a}{h}$$

a = assumed mean (pick a middle class mark), h = class size. All three methods give the same answer.

Mode of grouped data

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Modal class = class with the highest frequency. l = its lower limit, f_1 = its frequency, f_0 = frequency of the class before, f_2 = frequency of the class after, h = class size.

Median of grouped data

$$\text{Median} = l + \left(\frac{n/2 - cf}{f}\right) \times h$$

- n = Σf. Make a cumulative frequency (cf) column.
- Median class = first class whose cumulative frequency is ≥ $\frac{n}{2}$.
- l = lower limit of the median class, f = its frequency, **cf = cumulative frequency of the class *before* it**.

$$\text{Empirical relation: } 3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$$

Watch out: If classes are 10–19, 20–29 … (gaps), first make them continuous: 9.5–19.5, 19.5–29.5 … Needed for median and mode.

"Less than" / "more than" type tables: convert to ordinary class intervals before using the formulas.

Example · **Mean, median and mode from one table**

Class	f	x	f·x	u	f·u	cf
0–10	5	5	25	−2	−10	5
10–20	8	15	120	−1	−8	13
20–30	**15**	25	375	0	0	28
30–40	10	35	350	1	10	38
40–50	2	45	90	2	4	40
Total	**40**		**960**		**−4**	

$$\text{Mean (direct)} = \frac{960}{40} = 24$$

$$\text{Mean (step deviation, } a = 25, h = 10) = 25 + \left(\frac{-4}{40}\right) \times 10 = 24$$

Median: $\frac{n}{2} = 20$. First cf ≥ 20 is 28 ⇒ median class 20–30; l = 20, cf = 13, f = 15.

$$\text{Median} = 20 + \left(\frac{20 - 13}{15}\right) \times 10 \approx 24.67$$

Mode: modal class 20–30; $f_1 = 15$, $f_0 = 8$, $f_2 = 10$.

$$\text{Mode} = 20 + \left(\frac{15 - 8}{30 - 8 - 10}\right) \times 10 \approx 25.83$$

Check: 3 × 24.67 − 2 × 24 ≈ 26, close to the mode. The empirical relation is approximate.

Missing-frequency questions

1. Call the unknown frequencies x and y.
2. Equation 1: all frequencies add up to the given total n.
3. Equation 2: put the given mean (or median, or mode) into its formula.
4. Solve the two equations together.

For a given median, the median class is the class that contains the median value — use that to find l, f and cf in terms of x.

"Less than" table: "less than 10 → 5, less than 20 → 13, less than 30 → 28" means classes 0–10, 10–20, 20–30 with frequencies 5, 8, 15 (subtract successive values).

Which average to use

Measure	Best when	Weakness
Mean	every value should count (average marks)	pulled by extreme values
Median	typical / middle value (income, height)	ignores the actual size of other values
Mode	most popular item (shoe size, most-chosen option)	may not be unique

- Class size h = upper limit − lower limit. Class mark = mid-value.
- The mode formula needs equal class sizes; the mean and median formulas do not.
- Mean of grouped data assumes every value in a class sits at the class mark, so it is an estimate.

EXTRA **Ogive:** plot (upper limit, cf) for a "less than" ogive. The x-value at height $\frac{n}{2}$ is the median. Less-than and more-than ogives cross at the median.

14 · Probability

Rules

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of equally likely outcomes}}$$

- $0 \le P(E) \le 1$. Impossible event: 0. Sure event: 1.
- P(E) + P(not E) = 1.
- Sum of probabilities of all elementary events = 1.
- "At least one" = 1 − P(none).

Watch out: A probability can never be negative or greater than 1. Always reduce the fraction.

Sample spaces to remember

- **Coins:** 1 coin → 2 outcomes; 2 coins → 4 (HH, HT, TH, TT); 3 coins → 8.
- **One die:** 6 outcomes. Primes: 2, 3, 5.
- **Two dice:** 36 outcomes. Doublets: 6.

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

- **Cards (52):** 4 suits × 13. Red 26 (hearts, diamonds), black 26 (spades, clubs). Face cards 12 (J, Q, K × 4). Aces 4 — an ace is not a face card.
- **53 Sundays:** leap year $\frac{2}{7}$, ordinary year $\frac{1}{7}$.
- Numbers 1 to n: count carefully (multiples of 3 up to 50 → 16).

Vocabulary

- **Experiment / trial:** an action with an uncertain result. **Outcome:** one possible result. **Sample space:** the set of all outcomes.
- **Event:** a collection of outcomes. **Elementary event:** exactly one outcome.
- **Equally likely:** each outcome has the same chance. "Fair", "unbiased", "at random" and "well-shuffled" all signal this.
- **Complementary event (not E):** everything in the sample space outside E.
- "A **or** B": count outcomes in either, each only once. "A **and** B": count outcomes in both.
- Theoretical probability (this chapter) is calculated; experimental probability (Class 9) comes from actual trials.

Example · **Dice and coins**

Two dice (36 outcomes):

- P(sum 8) = $\frac{5}{36}$ · P(doublet) = $\frac{6}{36} = \frac{1}{6}$
- P(sum ≥ 10) = $\frac{3+2+1}{36} = \frac{1}{6}$
- P(product 6): (1,6), (6,1), (2,3), (3,2) ⇒ $\frac{4}{36} = \frac{1}{9}$
- P(sum is prime): sums 2, 3, 5, 7, 11 ⇒ $\frac{1+2+4+6+2}{36} = \frac{5}{12}$

Coins: two coins, at least one head = $\frac{3}{4}$. Three coins: exactly two heads = $\frac{3}{8}$; at most two heads = 1 − P(three heads) = $\frac{7}{8}$.

Example · **Cards and numbers**

One card from 52:

- P(red king) = $\frac{2}{52} = \frac{1}{26}$ · P(face card) = $\frac{12}{52} = \frac{3}{13}$
- P(spade or ace) = $\frac{13+3}{52} = \frac{4}{13}$ (the ace of spades counted once)
- P(neither king nor queen) = $\frac{52-8}{52} = \frac{11}{13}$
- If cards are removed first, reduce the total: all kings removed ⇒ 48 cards.

Card numbered 1 to 20: primes are 2, 3, 5, 7, 11, 13, 17, 19 ⇒ P(prime) = $\frac{8}{20} = \frac{2}{5}$.

Counting hints: two-digit numbers: 90 (10 to 99). Perfect squares from 1 to 100: 10. Numbers from a to b inclusive: b – a + 1.

Example · **Finding an unknown**

Bag with 5 red and x blue balls; P(blue) = 2 × P(red):

$$\frac{x}{5+x} = 2 \times \frac{5}{5+x} \Rightarrow x = 10$$

A box has 12 balls, x of them black. After adding 6 black balls the probability of black doubles:

$$\frac{x+6}{18} = 2 \times \frac{x}{12} \Rightarrow 12x + 72 = 36x \Rightarrow x = 3$$

Leap year with 53 Sundays: 366 days = 52 weeks + 2 extra days. The 7 possible pairs are (Sun, Mon), (Mon, Tue) … (Sat, Sun); 2 contain a Sunday ⇒ $\frac{2}{7}$.

Area type: P(point lands in a region) = $\frac{\text{area of the region}}{\text{total area}}$.

Defective items: P(good) = 1 – P(defective). Once an item is removed and not replaced, both the count and the total drop by 1.

Part C — Exam plan

Paper pattern (80 marks, 3 hours)

Section	Questions	Marks
A	18 MCQ + 2 assertion–reason, 1 mark each	20
B	5 very short answer, 2 marks each	10
C	6 short answer, 3 marks each	18
D	4 long answer, 5 marks each	20
E	3 case-study questions, 4 marks each	12

Plus 20 marks internal assessment. Confirm the current year's sample paper on cbseacademic.nic.in, since CBSE revises the pattern from time to time.

Unit weightage

Unit	Chapters	Marks
Number Systems	1	6
Algebra	2, 3, 4, 5	20
Coordinate Geometry	7	6
Geometry	6, 10	15
Trigonometry	8, 9	12
Mensuration	11, 12	10
Statistics and Probability	13, 14	11

Proofs to practise in full

- √2, √3, √5 are irrational.
- Basic Proportionality Theorem.
- Tangent is perpendicular to the radius at the point of contact.
- Tangents from an external point are equal.
- Trigonometric identity proofs (NCERT Exercise 8.3 style).

Mistakes that cost marks

- Sign errors in –b and in $b^2 - 4ac$ when c is negative.
- Using diameter where radius is needed.
- Taking cf of the median class itself instead of the class before it.
- Adding total surface areas in combined solids.
- Wrong order of vertices in similar triangles.
- Forgetting units (cm^2, cm^3) and the final statement in word problems.
- Not writing the formula first — the formula step earns a mark.
- Skipping the figure in geometry and height-and-distance questions.

How to prepare

- NCERT first: every example and exercise. Most board questions are NCERT or close variants.
- Then NCERT Exemplar for the harder MCQ and case-study types.
- Solve the last 5 years' board papers and the latest CBSE sample paper with a timer.
- Keep one formula notebook; revise it every week and the night before the exam.
- In the exam: read the paper for 15 minutes, do the sections you are sure about first, keep 10 minutes to recheck calculations.

Time plan for the 3 hours

Stage	Minutes	Note
Reading time	15 (given)	tick the questions you are sure of; choose between internal options
Section A (20 marks)	30	about 1.5 minutes each; do rough work, never guess blindly
Section B (10 marks)	20	2–3 steps each
Section C (18 marks)	40	formula → substitution → answer with unit
Section D (20 marks)	50	figure, given, to prove, full steps
Section E (12 marks)	25	read the case once fully, then answer the parts
Checking	15	signs, units, question numbers, anything left blank

Assertion–Reason questions

Decide separately whether the Assertion (A) and the Reason (R) are true. Then:

Option	Meaning
(a)	A and R both true, and R correctly explains A
(b)	A and R both true, but R does not explain A
(c)	A true, R false
(d)	A false, R true

Test for (a) versus (b): can you say "A is true *because* R"? If that sentence makes sense, choose (a).

Must-do question types, chapter by chapter

Ch	Practise until automatic
1	HCF–LCM word problem; irrationality proof
2	zeros + verify; build polynomial; expressions in α, β
3	find k for each case; one word problem of each type; graph
4	nature of roots; find k; speed and work problems
5	find n; sum with last term; S_n given; real-life AP
6	BPT proof; proving triangles similar; shadow problems
7	ratio by an axis; fourth vertex; equidistant point
8	value-based evaluation; 6 to 8 identity proofs
9	two-triangle problems of each set-up
10	both proofs; circumscribed quadrilateral and triangle
11	segment area for 60°, 90°, 120°; shaded regions
12	one of each: combination, recasting, flow
13	mean (all 3 methods), median, mode; missing frequency
14	two dice; cards; find-x problems

Presentation that earns full marks

- Write the question number clearly and leave a line between answers.
- Step marking: formula (1 mark), substitution, calculation, final answer with unit. A wrong final answer still earns step marks.
- Word problems: start with "Let … be x" and end with a sentence that answers the question asked.
- Proofs: Given, To prove, Construction, Proof, with a reason in brackets beside each step.
- Draw figures with pencil and ruler, label every point used in the solution.
- Box or underline the final answer. Keep rough work in a margin column, not on scattered scraps.
- Leave answers as fractions or surds unless told to use decimals; rationalise denominators.

Case-study questions (Section E)

- Each has a real-life passage and usually three parts worth 1, 1 and 2 marks, with a choice in the 2-mark part.
- First decide which chapter it is: seats in rows → AP; tower and angles → heights and distances; grid positions → coordinate geometry; tent or toy → surface areas and volumes; frequency table → statistics.
- Pull out the numbers and write them as a, d, r, h, θ … before reading the sub-questions.
- The parts are usually independent, so a mistake in one need not spoil the others.

Last 30 days

- **Days 1–14:** one chapter a day — read this sheet's chapter, redo the NCERT examples, then 15 mixed questions.
- **Days 15–24:** one full sample or past paper every day under exam timing; mark it against the official marking scheme.
- **Days 25–28:** redo only the questions you got wrong; keep an error log by type (concept, calculation, misreading).
- **Days 29–30:** formulas, proofs and the standard-values table only. Sleep properly.

Standard or Basic: CBSE offers two papers on the same syllabus. Standard (code 041) is the harder paper and the usual route to Mathematics in Class 11; Basic (code 241) is for students not continuing with maths. Check the current rule with the school before choosing.